

Instability identifies structured regulatory signals beyond condition-level differential expression

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Biological responses to the same experimental condition often vary across temporal phases, biological subgroups, or response regimes, creating structured internal response diversity that is not fully represented by a single condition-level differential expression analysis. Although conventional differential expression effectively identifies dominant transcriptional responses, it does not explicitly quantify the consistency of gene regulation across these internal biological partitions. We introduce **instability**, a quantitative measure that captures internal response disagreement and provides an additional analytical dimension for interpreting differential expression in heterogeneous biological conditions. Using synthetic benchmarks, we demonstrate that instability selectively increases under opposing and temporally structured regulatory regimes while remaining low in coherent and homogeneous conditions. Across real transcriptomic datasets representing ultraviolet irradiation, desiccation, gamma irradiation, and low-temperature stress, instability was consistently elevated in biologically heterogeneous conditions and remained minimal in homogeneous responses. Instability was independent of expression variance and generated gene prioritizations that differed substantially from condition-level differential expression, identifying biologically meaningful genes with structured subgroup- or time-dependent regulation that received comparatively little emphasis under conventional condition-level summaries. These included regulatory and neuroendocrine genes associated with distinct biological response programs. To facilitate reproducible analysis, we implemented the framework as an open-source computational pipeline compatible with standard transcriptomic workflows. By quantifying internal response diversity rather than only dominant condition-level effects, instability extends the interpretation of differential expression and provides a complementary framework for identifying structured regulatory signals in heterogeneous biological datasets.

Keywords: Differential expression, Transcriptomics, Gene prioritization, Internal response diversity, Instability, RNA-seq, Computational biology.

1. Introduction

Biological responses to the same experimental condition are often not internally uniform. Temporal progression, biological subgroups, physiological states, and other structured biological partitions can produce distinct regulatory programs within a single experimental condition. Such internal response diversity is common in longitudinal, perturbation, and large-scale transcriptomic studies and complicates the interpretation of condition-level differential expression (Love, et al., 2014; Ritchie, et al., 2015; Love, et al., 2015; Soneson and Robinson, 2018). Quantifying differential gene expression under heterogeneous biological conditions remains an unresolved challenge despite the maturity of transcriptomic analysis pipelines. Standard differential-expression workflows summarize gene-level responses by estimating a single effect for each experimental condition using all samples assigned to that condition. Throughout this study, we refer to this estimated transcriptional response as the **condition-level differential expression**, distinguishing it from physical pooling of biological samples. Consequently, the resulting condition-level estimate represents the dominant transcriptional response of the experimental condition while not explicitly characterizing variation across internal biological partitions. This analytical strategy underlies widely used frameworks such as DESeq2 and limma (Love, et al., 2014; Ritchie, et al., 2015), as well as established RNA-seq analysis and benchmarking practices (Love, et al., 2015; Soneson and Robinson, 2018).

This condition-level representation becomes insufficient when responses within an experimental condition are not internally coherent. In the presence of subgroup-specific or time-dependent regulation, genes may exhibit responses that differ in both magnitude and direction across biological partitions (Fig. 1a). Under these conditions, a single condition-level summary no longer represents the underlying regulatory structure because opposing or divergent responses become attenuated when summarized into a single condition-level effect (Fig. 1b). As illustrated in Fig. 1, these structured internal responses can converge toward an overall effect close to zero, creating the appearance of weak or absent regulation despite strong underlying biological activity (Fig. 1c). This represents a systematic limitation in which structured regulatory signals become underrepresented in condition-level summaries, consistent with observations in heterogeneous biological systems and mixed cell populations (Newman, et al., 2015; Squair, et al., 2021; Tsoucas and Yuan, 2018; Wagner, et al., 2016). Because conventional differential-expression analyses do not explicitly quantify the consistency of responses across biological partitions, such structured internal disagreement can remain difficult to recognize using standard differential-expression workflows alone.

Existing approaches for analyzing heterogeneous biological responses extend conventional differential-expression models through subgroup analyses, interaction terms, or time-course designs (Bar-Joseph, et al., 2012; Conesa, et al., 2016; Spies, et al., 2019; Storey, et al., 2005). These approaches effectively identify differential expression within predefined biological groups or temporal relationships when the experimental structure is known. However, they are not designed to directly quantify the consistency of gene regulation across biological partitions within

the same experimental condition. Consequently, while individual subgroup- or time-specific effects can be characterized, the degree to which these responses are mutually consistent or divergent across partitions remains unquantified. This limitation is particularly relevant in complex or dynamic biological systems, including longitudinal, single-cell, and pseudobulk transcriptomic studies, where internal response diversity may be only partially characterized or evolve over time (Amezquita, et al., 2020; Luecken and Theis, 2019; Stuart and Satija, 2019; Vallejos, et al., 2015).

Variance-based measures do not resolve this limitation. Although variance captures the overall dispersion of gene expression, it does not distinguish between random variability and structured divergence arising from conflicting responses across biological partitions (Hicks, et al., 2018; Leek, et al., 2010; McCarthy, et al., 2012; Risso, et al., 2014). Consequently, variance cannot differentiate stochastic noise from biologically meaningful internal response disagreement. As a result, current analytical frameworks lack a quantitative measure of the consistency of gene regulation across biological partitions within the same experimental condition.

To address this limitation, we introduce **instability** as a quantitative measure of internal response diversity that captures both differences in response magnitude and direction across biological partitions. Unlike conventional differential-expression frameworks, which summarize each gene by a single condition-level effect, instability explicitly quantifies the consistency of gene regulation across predefined biological partitions within the same experimental condition. By jointly evaluating partition-specific responses, instability characterizes structured regulatory behavior that is not captured by condition-level differential expression alone.

Treating instability as a complementary analytical dimension extends the interpretation of heterogeneous transcriptomic data. Genes with modest condition-level differential expression may exhibit high instability, indicating structured regulatory divergence across biological partitions rather than the absence of biological signal. Conversely, genes with strong condition-level differential expression and low instability represent coherent regulatory responses that are consistently maintained across biological partitions. Together, condition-level differential expression and instability provide complementary perspectives on transcriptional regulation by distinguishing dominant responses from internally divergent regulatory programs.

In this study, **heterogeneity** refers to structured internal response diversity within an experimental condition rather than differences between treatment and control groups. We define **instability** as a quantitative measure of the consistency of gene regulation across predefined biological partitions and use it to identify structured regulatory signals that are not adequately represented by condition-level differential expression alone. Using controlled synthetic benchmarks and real stress-response transcriptomic datasets, we demonstrate that instability captures a distinct dimension of biological response that is independent of expression variance and reveals biologically meaningful patterns of internal response diversity in heterogeneous biological systems (Consortium, 2014; Kelsey, et al., 2017; Li, et al., 2014; Trapnell, et al., 2014).

2. Methods

2.1. Data sources and transcriptomic preprocessing

RNA-seq datasets were derived from stress-response experiments in the tardigrade *Ramazzottius varieornatus*, a model organism known for extreme tolerance to environmental stress. Publicly available SRA datasets included ultraviolet exposure (UV), desiccation (DES), gamma radiation (GAM), and low-temperature conditions (LT). Each dataset contained structured internal partitions defined either by biological subgroups (e.g., UV short vs long exposure; DES fast vs slow dehydration) or by temporal progression (GAM timepoints). LT was used as a homogeneous reference condition. Additional stress conditions, including high temperature (HT) and osmotic stress (OSM), are present in the dataset but are not central to the analyses used to demonstrate the instability framework, which focuses on conditions with well-defined internal structure.

Metadata were standardized using a custom preprocessing pipeline to harmonize sample identifiers, condition labels, treatment status, and partition structure (Supplementary Table S1) (Love, et al., 2015; Sonesson and Robinson, 2018). The resulting curated metadata table was used consistently for all condition-level and partition-specific analyses.

2.2. Definition of biological partitions and response structure

A condition was defined as homogeneous when gene expression responses were consistent in both direction and magnitude across its internal samples (e.g., timepoints or replicates), such that condition-level summaries closely approximated subgroup-aware analyses.

A condition was considered to exhibit structured internal response diversity when predefined biological partitions showed systematic differences in transcriptional response magnitude or direction. Such conditions were analyzed using partition-specific differential-expression estimates to quantify response consistency across biological partitions.

Importantly, this definition of heterogeneity is independent of the presence of treatment and control groups. Each condition was analyzed relative to its own baseline control, corresponding to nonstressed samples (e.g., 0h timepoints or untreated controls within the same experiment). Although these controls represent a common biological baseline, they are defined independently within each dataset. The internal structure of each condition was then evaluated based on the consistency of gene expression responses across its samples. Conditions with consistent direction and magnitude were considered homogeneous, whereas those showing subgroup-specific or time-dependent divergence were classified as exhibiting structured internal response diversity.

2.3. RNA-seq quantification and expression matrix construction

Transcript-level quantification was performed using Salmon. Gene-level expression estimates were obtained using tximport with a transcript-to-gene mapping (tx2gene.tsv) (Sonesson, et al.,

2016), followed by DESeq2 variance-stabilizing transformation (VST) (Love, et al., 2014) using default parameters (blind = FALSE). Principal component analysis (PCA) was performed on the VST-transformed matrix for exploratory analysis and visualization using the prcomp function in R.

All downstream analyses were performed on this gene-level VST matrix. Hereafter, gene-level expression estimates are referred to simply as genes for clarity.

2.4. Condition-level and partition-specific effect estimation

Condition-level differential expression was estimated using a linear modeling framework implemented in **limma**, in which all samples assigned to an experimental condition were analyzed jointly to estimate a single treatment-control effect for each gene. Partition-specific differential-expression estimates were obtained using the same framework, but treatment-control contrasts were calculated independently within predefined biological partitions, including biological subgroups or temporal phases.

For each gene i , differential expression was estimated as the $\log_2$ fold change ($\log_2\text{FC}$). Let $\Delta_{i,k}$ denote the partition-specific $\log_2$ fold change for gene i in biological partition k , where K is the total number of predefined partitions.

The condition-level effect, $\Delta_{\text{pool},i}$ (where **pool** denotes the **condition-level** differential-expression estimate), was defined as the treatment-control $\log_2$ fold change estimated from the condition-level differential-expression model using all samples assigned to the experimental condition. This estimate represents the overall treatment-control response for each gene i , but does not retain partition-specific response structure.

For descriptive comparison across biological partitions, the mean partition-specific effect for gene i was defined as:

$$\bar{\Delta}_i = \frac{1}{K} \sum_{k=1}^K \Delta_{i,k}$$

where K denotes the number of biological partitions.

Partition-specific effects $\Delta_{i,k}$ were used to quantify response divergence across biological partitions, whereas the condition-level effect $\Delta_{\text{pool},i}$ was used for masked-gene identification and condition-level gene-ranking analyses.

2.5. Instability score

Instability was defined to quantify internal response disagreement across predefined biological partitions by combining effect divergence and directional inconsistency.

Effect divergence was defined as:

- **for subgroup-based conditions:**

$$D_i = | \Delta_{i,1} - \Delta_{i,2} |$$

- **for temporal conditions:**

$$D_i = \max_t (\Delta_{i,t}) - \min_t (\Delta_{i,t})$$

Directional discordance was defined as disagreement in the direction of differential expression across biological partitions. For subgroup-based conditions:

$$C_i = \begin{cases} 1, & \text{if } \text{sign}(\Delta_{i,1}) \neq \text{sign}(\Delta_{i,2}) \\ 0, & \text{otherwise} \end{cases}$$

For temporal conditions, directional discordance was defined as the fraction of timepoints whose direction of differential expression differed from the dominant response direction.

The instability score was then calculated as:

$$I_i = D_i \cdot (1 + C_i)$$

This formulation combines response magnitude divergence and directional inconsistency, representing the two principal components of structured internal response diversity across biological partitions. Consequently, instability increases with both increasing response divergence and increasing directional disagreement.

2.6. Definition of masked genes

Genes were classified as masked when weak condition-level differential expression coincided with high instability. Let $Q_{0.25}$ denote the 25th percentile of $| \Delta_{pool,i} |$, and $Q_{0.90}$ the 90th percentile of instability. A gene i was defined as masked if:

$$| \Delta_{pool,i} | \leq Q_{0.25} \text{ and } I_i \geq Q_{0.90}$$

The masked fraction within each condition was computed as the proportion of genes satisfying these criteria.

Thresholds were defined using fixed quantile cutoffs to identify regions of low condition-level effect and high instability. Robustness of instability-based prioritization was assessed across a range of selection thresholds (Top-N) (Fig. 5B).

2.7. Variance proxy and regression analysis

To assess whether instability reflects dispersion alone, a variance proxy was computed from structured effect estimates as the variance of $\Delta_{i,k}$ across partitions (Leek, et al., 2010; McCarthy, et al., 2012).

The relationship between masking and instability was evaluated using logistic regression (Lemeshow and Hosmer, 2009):

$$\text{logit}(P(M_i = 1)) = \beta_0 + \beta_1 I_i + \beta_2 V_i$$

where M_i indicates masked genes, I_i is instability, and V_i is the variance proxy.

2.8. Cross-condition analysis

Instability distributions were compared across UV, DES, GAM, and LT conditions. Enrichment of high-instability genes was quantified using empirical thresholds, including the fractions of genes exceeding the top 10% and top 5% instability levels.

Instability was further decomposed into its components (effect divergence and directional discordance) to characterize condition-specific contributions.

2.9. Gene prioritization and overlap analysis

Genes were ranked independently according to condition-level effect magnitude and instability:

$$R_i^{pool} = \text{rank}(\Delta_{pool,i}), R_i^{inst} = \text{rank}(I_i)$$

Top-N gene sets were defined for each ranking scheme:

$$S_N^{pool} = \{i: R_i^{pool} \leq N\}, S_N^{inst} = \{i: R_i^{inst} \leq N\}$$

Instability-only and condition-level-only sets were defined as set differences:

$$S_N^{inst-only} = S_N^{inst} \setminus S_N^{pool}, S_N^{pool-only} = S_N^{pool} \setminus S_N^{inst}$$

Overlap between sets was quantified using the Jaccard index:

$$J(N) = \frac{|S_N^{pool} \cap S_N^{inst}|}{|S_N^{pool} \cup S_N^{inst}|}$$

Analyses were performed for $N = 50, 100, 200, \text{ and } 500$.

Representative genes were selected from instability-only and condition-level-only sets based on objective criteria, including (i) rank displacement defined as $|R_i^{pool} - R_i^{inst}|$, (ii) instability score magnitude, and (iii) consistency of structured effect patterns, defined as absence of stochastic

fluctuations across partitions relative to the dominant trend. Selection was restricted to genes within the Top-N ranked sets used for overlap analysis, ensuring consistency with the global prioritization framework. No manual curation based on gene identity or prior functional knowledge was applied.

Functional annotation was assigned post hoc by mapping selected gene identifiers to reference genome annotations and associated protein records, followed by integration of annotations from NCBI, eggNOG, InterProScan, and Pfam databases.

2.10. Synthetic benchmarking and variance disentanglement

Synthetic datasets were constructed to represent coherent, opposing, and phase-structured regulatory regimes. For each scenario, condition-level effect, instability, and variance proxy were computed as described above.

Summary statistics were reported as mean $\pm$ standard deviation or median and interquartile range. To assess whether instability is reducible to variance, changes in instability and variance were compared across scenarios.

2.11. Robustness and uncertainty estimation

Threshold robustness was evaluated across ranges of condition-level effect and instability quantiles. Stability of condition separation and gene prioritization was assessed across these parameter settings.

Uncertainty in masked fractions and related statistics was estimated using bootstrap resampling, and results were summarized as mean $\pm$ standard deviation.

2.12. Software implementation

All analyses were performed in **R v4.4.3** using a custom analysis pipeline developed for instability quantification and downstream transcriptomic analysis. Transcript quantification was performed using **Salmon v1.10.2**, followed by gene-level summarization using **tximport v1.34.0** and variance-stabilizing transformation using **DESeq2 v1.46.0**. Differential-expression analyses were performed using **limma v3.62.1**, and all downstream statistical analyses, visualization, and instability calculations were implemented in R.

3. Results

To evaluate instability across distinct forms of internal heterogeneity, we analyzed RNA-seq datasets representing subgroup- and time-dependent stress responses in *Ramazzottius varieornatus*. Ultraviolet exposure (UV) contained temporally distinct response regimes with partially conflicting effects, desiccation (DES) exhibited subgroup-dependent response magnitude differences, and gamma radiation (GAM) represented temporal divergence across stress progression. In contrast, low-temperature controls (LT) showed comparatively consistent

responses and were used as a relatively homogeneous reference condition. Here, heterogeneity refers to structured internal variation within a condition, such as temporal phases or biological subgroups, rather than differences between treatment and control groups.

3.1. Pooled analysis attenuates structured signals across synthetic regulatory regimes

We evaluated pooled analysis under controlled synthetic regimes (Fig. 2a–c). Pooled estimates preserved signal in coherent scenarios (pooled: 1.101 ± 0.036 ; structured: 1.102 ± 0.036), but strongly attenuated structured signals. In opposing scenarios, mean absolute effect decreased from 1.497 ± 0.019 (structured) to 0.283 ± 0.019 (pooled), a ~ 5.3 -fold reduction. In phase-structured time-course data, pooled magnitude (0.251 ± 0.022) was reduced relative to structured estimates (0.850 ± 0.012), a ~ 3.4 -fold decrease, while coherent signals remained stable (pooled: 0.999 ± 0.031 ; structured: 1.001 ± 0.031). Null scenarios showed low effects under both analyses (subgroup: 0.282 ± 0.008 vs 0.325 ± 0.007 ; time-course: 0.252 ± 0.006 vs 0.325 ± 0.006).

Masking was negligible in coherent and null regimes but substantial in structured cases, with masked fractions of 0.278 ± 0.020 (opposing) and 0.281 ± 0.025 (phase), indicating that approximately 28% of genes were masked under structured regimes, defined by low-to-moderate pooled effect and high instability.

These results demonstrate that attenuation under pooling is not a consequence of noise or insufficient signal strength but arises specifically from structured disagreement between underlying responses. In such cases, pooled estimates provide no indication that biologically meaningful effects have been suppressed.

3.2. Instability is distinct from variance and reveals masked signals

To assess whether instability reveals signals not captured by pooled analysis, we compared pooled effect size and instability across 15,187 genes in the UV dataset (Fig. 3a–d). Using thresholds $Q0.25(|\Delta_{\text{pool}}|) = 0.119$ and $Q0.90(\text{instability}) = 1.146$, 283 genes ($\sim 1.9\%$) were identified within a masked region characterized by low pooled effect but high instability. Representative genes (e.g., RvY_14730-1, RvY_14133, RvY_06540-1; Supplementary Table S2) showed strong opposing subgroup effects ($|\Delta| \approx 1.18\text{--}1.31$) that collapsed toward near-zero when pooled.

Logistic regression confirmed instability as a strong predictor of masking ($\beta = 3.16 \pm 0.13$, $z = 24.28$, $p < 2 \times 10^{-16}$), whereas variance showed a weaker negative association ($\beta = -0.65 \pm 0.23$, $z = -2.86$, $p = 0.004$), improving overall model fit ($\Delta\text{deviance} = 847.8$; $\text{AIC} = 1973.1$). This supports a distinction between variance as a measure of dispersion and instability as a measure of structured divergence.

Grouping genes by instability revealed a progressive increase in directional discordance. Median divergence rose from 0.40 (IQR: 0.18–0.72) in the full set to 1.33 (1.19–1.55) and 1.55 (1.42–1.77) in the top 10% and 5% instability tiers, respectively (~ 3.3 -fold increase). Masked genes

consistently exhibited opposing subgroup effects, indicating structured cancellation rather than weak signal. DES-specific patterns are shown in Supplementary Fig. S1.

3.3. Instability differs across structured and homogeneous conditions

Instability distributions varied across conditions (Fig. 4a), with median values of 0.43 (UV), 0.39 (DES), and 0.77 (GAM), compared with 0.21 in the homogeneous LT condition. Structured conditions showed broader distributions and extended high-instability tails, whereas LT remained tightly constrained at low values, consistent with a homogeneous response.

Consistent with this pattern, high-instability tail occupancy was enriched in structured conditions (Fig. 4b). The fraction of genes within the top 10% instability range reached 10% (UV), 9% (DES), and 20% (GAM), compared with 0.8% in LT. Similarly, the top 5% tail showed enrichment in UV (4.8%), DES (5.3%), and GAM (9.6%), but remained minimal in LT (0.3%).

Decomposition of instability into heterogeneity gap and directional discordance revealed distinct condition-specific drivers (Fig. 4c). GAM exhibited the highest heterogeneity (0.67) and discordance (0.125), indicating strong temporal divergence. DES showed comparable heterogeneity (0.39) but negligible discordance, consistent with magnitude differences across subgroups. UV displayed intermediate heterogeneity (0.40) with modest discordance (0.026), reflecting partial subgroup conflict. In contrast, LT exhibited uniformly low heterogeneity (0.21) and near-zero discordance.

These structural differences translated into condition-specific masking behavior (Fig. 4d). The fraction of masked genes remained minimal in LT (0.0066%) but increased in structured conditions (UV 1.40%, DES 0.34%, GAM 3.37%; bootstrap means: 0.014 ± 0.002 , 0.004 ± 0.001 , 0.034 ± 0.003).

Together, these results show that instability is selectively elevated in the presence of structured internal variation and remains low in homogeneous conditions, indicating that it captures structured disagreement rather than overall signal magnitude.

3.4. Instability identifies genes missed by pooled analysis

Across all genes ($n = 15,187$), instability-based prioritization diverged strongly from pooled effect-based ranking (Fig. 5a). Rank-rank comparison revealed a distinct subset of genes that were highly prioritized by instability but ranked poorly by pooled log2 fold change, with representative genes (RvY_14730-1, RvY_18684-1, RvY_14091-1) shifting by more than 11,000 rank positions. This displacement indicates systematic masking of structured responses under aggregation.

Consistent with this displacement, instability- and pooled-based selections showed minimal overlap (Fig. 5b). At $N = 100$, no shared genes were observed, indicating distinct prioritization regimes. This separation reflects fundamentally different prioritization criteria rather than

instability of ranking, with instability selectively capturing genes exhibiting structured heterogeneity, in contrast to pooled analysis, which favors genes with consistent average effects.

The separation between prioritization regimes remained stable across selection thresholds (Fig. 5c). Overlap remained negligible at $N = 50$ and $N = 100$, and increased only modestly at larger thresholds (Jaccard index $J(200) = 0.020$; $J(500) = 0.046$), corresponding to 8 and 44 shared genes, respectively, and remaining below 5% even under relaxed thresholds.

Representative examples illustrate the contrasting biological patterns captured by each approach (Fig. 5d). Instability-selected genes showed high instability (2.89–2.98) and strong effect divergence across groups (2.75–2.85), consistent with opposing or subgroup-dependent responses. In contrast, pooled-selected genes exhibited low instability (0.08–0.28) and consistent regulation across groups, reflecting stable average responses.

Functional characterization of instability-recovered masked genes further supports their biological relevance (Fig. 5e). Domain-based annotation (Pfam, InterProScan) revealed representation across multiple functional classes, including signaling/receptor proteins, metabolic enzymes, and nucleic acid-associated regulators, with complete annotation and classification provided in Supplementary Table S3. Notably, instability-based prioritization recovered neuroendocrine signaling peptides, including Corazonin (Crz), Pigment-Dispersing Hormone (PDH), and Crustacean Cardioactive Peptide (CCAP), which were not prominent under pooled analysis (Supplementary Table S3).

3.5. Instability reveals structured regulatory patterns in heterogeneous data

To test whether instability captures structured disagreement beyond variance, we performed controlled synthetic analysis across defined response regimes (Fig. 6). Median instability rose from 0.277 (coherent) to 1.815 (phase) and 3.410 (opposing), whereas variance increased from 0.034 to 0.506 and 2.292, and pooled effects decreased from 1.042 to 0.143 and 0.164.

Instability increased ~6.5-fold (phase) and ~12-fold (opposing), whereas variance increased ~67-fold, indicating non-proportional scaling. Variance reflects dispersion, whereas instability captures structured divergence across groups. Importantly, the non-proportional and scenario-dependent scaling indicates that instability is not a monotonic transformation of variance.

Together, these results show that instability quantifies internal disagreement rather than variance magnitude, providing a distinct signal for heterogeneous regulatory regimes.

4. Discussion

In this study, we demonstrate that structured internal response diversity is an intrinsic property of many heterogeneous transcriptomic datasets and should be explicitly considered when interpreting condition-level differential expression. When regulatory responses differ across predefined biological partitions, a single condition-level transcriptional summary can attenuate structured regulatory signals, reducing opposing responses toward near-zero estimates and leading to an

incomplete representation of the underlying biological response (Newman, et al., 2015; Squair, et al., 2021; Tsoucas and Yuan, 2018; Wagner, et al., 2016).

Instability provides a quantitative measure of this internal disagreement across predefined biological partitions. Across synthetic and real datasets, instability increased selectively under opposing and phase-structured regimes while remaining low in coherent and null conditions. In real data, instability identified genes with subgroup-dependent or temporally structured responses that were not captured by condition-level summaries. Unlike subgroup-specific analyses, which recover individual effects but do not quantify their compatibility, instability provides a unified measure of whether responses are coherent or conflicting across groups.

A defining property of instability is its condition-specific activation. It is elevated only when structured internal response diversity is present (UV, DES, GAM) and remains minimal in homogeneous conditions (LT). This behavior differentiates instability from variance, which reflects dispersion but cannot differentiate stochastic variability from structured disagreement (Leek, et al., 2010; McCarthy, et al., 2012). Decomposition analysis further shows that instability arises from distinct structural drivers, including directional conflict, magnitude divergence, and temporal variation, supporting its interpretation as a structure-sensitive measure rather than a transformation of variance. Temporal regimes such as GAM may partially couple instability with response amplitude variation, indicating that some forms of temporal divergence can contribute variance-associated structure alongside directional disagreement (Supplementary Fig. S2).

A direct consequence of this behavior is divergence in gene prioritization. Instability-based ranking identifies genes largely distinct from those selected by condition-level differential-expression ranking, reflecting fundamentally different prioritization criteria rather than instability of ranking. While condition-level differential expression preferentially selects genes with consistent directionality, instability highlights genes characterized by structured, subgroup-dependent, or temporally heterogeneous responses. As a result, genes that appear weak under condition-level differential expression may represent structured regulation that becomes detectable only through instability.

In conventional analysis workflows, such genes often require extensive iterative subgroup-specific analyses, including subgroup-specific contrasts or time-resolved analyses, to resolve underlying structure. Instability provides a direct measure of internal disagreement, enabling their identification without prior stratification or iterative contrast testing, thereby reducing reliance on ad hoc post hoc analysis.

The functional composition of the masked gene set further supports its biological relevance. Domain-based annotation of all masked genes (Supplementary Table S3) revealed multiple regulatory and signaling-associated features, including kinase/phosphatase modules, GPCR-related domains, and nucleic acid-associated components, consistent with stress-response pathways. Notably, instability-based prioritization recovered neuroendocrine signaling peptides,

including Corazonin (Crz), Pigment-Dispersing Hormone (PDH), and Crustacean Cardioactive Peptide (CCAP), which were not prominent under condition-level differential expression. These genes were identified within the masked set prior to functional interpretation, indicating that their prioritization was driven by structured instability rather than annotation completeness. Given that such signaling molecules are often expressed in spatially restricted cell populations or exhibit temporally dynamic expression, their signals are susceptible to attenuation under pooled transcriptomic analysis. Their recovery is therefore consistent with structured regulatory heterogeneity that is not captured by condition-level summaries.

Instability quantifies structured disagreement of effect across defined groups, enabling the identification of genes whose regulatory behavior cannot be captured by a single condition-level estimate. Rather than replacing differential expression analysis, it complements it by identifying when condition-level summaries are incomplete or potentially misleading. In heterogeneous datasets, condition-level estimates can obscure structured regulatory patterns, leading to incorrect biological interpretation rather than reduced sensitivity. While subgroup-specific analyses can recover individual effects, they require prior knowledge of relevant structure and do not provide a unified measure of compatibility across groups. Instability addresses this gap by quantifying whether a single effect estimate is meaningful.

A limitation of the current framework is its reliance on predefined subgroup or temporal groups. In settings where structure is latent, additional approaches may be required to identify relevant groups before instability can be applied, as the method does not infer the source of heterogeneity but instead quantifies its presence and structure (Bar-Joseph, et al., 2012; Conesa, et al., 2016; Spies, et al., 2019; Storey, et al., 2005). Consequently, instability complements existing analyses by quantifying response consistency across known biological partitions rather than identifying the source of biological heterogeneity itself (Bar-Joseph, et al., 2012; Conesa, et al., 2016; Spies, et al., 2019; Storey, et al., 2005).

The applicability of instability extends to transcriptomic analyses where biologically meaningful partitions are encoded in the experimental design, including subgroup, temporal, developmental, perturbation, or condition-specific analyses. Consequently, its utility is expected to be greatest in heterogeneous biological systems where internal response diversity is biologically meaningful rather than technical in origin.

Structured internal response diversity is a common characteristic of many biological systems, including stress adaptation, development, disease progression, and heterogeneous cell populations (Consortium, 2014; Kelsey, et al., 2017; Li, et al., 2014; Trapnell, et al., 2014). By quantifying the consistency of gene regulation across predefined biological partitions, instability extends the interpretation of condition-level differential expression and provides a complementary analytical framework for identifying biologically meaningful regulatory programs in heterogeneous transcriptomic datasets.

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Competing Interests

The authors declare no competing interests.

Code and Data Availability

All analyses were performed using custom, fully documented R scripts. The complete analytical pipeline, including preprocessing, effect estimation, instability calculation, supplementary analyses, and figure generation, is maintained in a version-controlled GitHub repository (https://github.com/sajadshahbaz/NM2_instability_signal). The repository is currently private during peer review and will be made publicly available upon acceptance of the manuscript. A permanent archived Zenodo release and DOI will be generated at the time of public release. Processed datasets, metadata tables, precomputed resources required for figure generation, and intermediate outputs used to reproduce the manuscript figures are maintained within the repository. Raw RNA-seq datasets are publicly available through the Sequence Read Archive (SRA) under the BioProject accessions listed in Supplementary Table S1, comprising 111 samples across six independent studies. Metadata used for condition definition, partitioning, and downstream analyses are provided in Supplementary Table S1.

All results reported in this study can be reproduced using the supplied scripts, processed inputs, and default parameters. Computational environment specifications, software dependencies, and reproducibility instructions are documented within the repository. Repository access can be provided to editors and reviewers upon request during the peer-review process.

Author Contributions

S.S.: Conceptualization; Methodology; Data curation; Formal analysis; Investigation; Validation; Visualization; Writing – original draft, Writing – review & editing. **P.M.:** Funding acquisition; Project administration; Resources; Supervision; Writing – review & editing. **J.L.:** Investigation; Validation; Figure preparation; Writing – review & editing. **A.U.:** Scientific guidance; Writing – review & editing. **T.Z.:** Scientific guidance; Writing – review & editing.

Use of Generative AI Tools

Generative artificial intelligence (AI) tools, including OpenAI ChatGPT, Rubriq, and American Journal Experts (AJE) language-editing services, were used during manuscript preparation to assist with English language editing, structural refinement, code refinement, and the development of preliminary conceptual layouts for schematic figures, including the conceptual workflow and graphical abstract. These tools were used solely as assistive aids and not for the design of the

methodological framework, mathematical formulation, algorithm implementation, parameter selection, data processing, statistical analyses, or interpretation of results. All scientific concepts, analytical decisions, figures, and conclusions were developed, verified, and approved by the authors, who take full responsibility for the accuracy, originality, and integrity of the manuscript.

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